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**Experiment 04**

**Aim:** To understand continuous integration,

Jenkins installation

This experiment maps to LO1,LO3:

LO1:Apply Agile methodologies and use JIRA for sprint planning, task tracking and integrating DevOp workflows

LO3:Build and Manage automated CI/CD Pipeline for application development, testing, packaging, and deployment.

Theory:

Jenkins is an open-source automation server used for implementing Continuous Integration (CI) and Continuous Delivery/Deployment (CD). It automates tasks such as building code, running tests, generating reports, packaging applications, and deploying software. Jenkins supports integration with tools like Git, Maven, Gradle, JUnit, and cloud platforms through its large plugin ecosystem. It is developed in Java and runs on the JVM, providing a web-based interface to manage projects, pipelines, users, and plugins.

A Jenkins Pipeline is a set of automated steps that defines the software delivery process. It is written in a file called Jenkinsfile, which stores pipeline instructions as code. A typical pipeline includes stages like:

- Build – Compiles source code and creates the application.
- Test – Runs automated tests to check quality and functionality.
- Deploy – Releases the application to servers or production.

Advantages of Jenkins Pipeline

- Automates software delivery.
- Reduces manual errors.
- Speeds up development and testing.
- Provides better visibility of the build process.
- Supports continuous integration and deployment.
- Allows pipeline configuration to be stored and shared as code.

Types of Jenkins Pipelines

1. Declarative Pipeline

- Uses the pipeline {} block.
- Has a simple, fixed structure.
- Easier to read and maintain.
- Suitable for beginners.

Example:

```
pipeline {
    agent any
    stages {
        stage('Build') {
            steps {
                echo 'Building application'
            }
        }
    }
}
```

2. Scripted Pipeline
   - Written using Groovy scripting.
   - Provides more flexibility and control.
   - Suitable for complex workflows.
   - Requires more programming knowledge.

Example:

```
node {
    stage('Build') {
        echo 'Building application'
    }
}
```

Jenkins plays an important role in DevOps by automating the software development lifecycle, improving efficiency, reducing errors, and enabling faster and more reliable software delivery.

Implementation

1. Check if JDK (Java Development Kit) is installed.

```
C:\Users\15L>javac -versio
javac 21.0.2
```

2. Download Jenkins WAR file from official website or packet manager

**jenkins.war**

permalink to the latest

| | |
|---|---|
| 2.575 | Released: Jul 28, 2026<br>SHA-1: 781df4433fa03537c429df8b909d2f3ebeb0d8d3<br>SHA-256: 183c18356b76b6680c7d2f689258acc8899aff6c7573d6a0e3ae0852f85b2bfc |
| 2.574 | Released: Jul 21, 2026<br>SHA-1: 143d74dfc83f5f124c5b6a5ac262b1788c1ba483<br>SHA-256: e924a55dff7e96efed4844df55c6433b8dc03c7e639b45f8289e6df8b893fca2 |
| 2.573 | Released: Jul 14, 2026<br>SHA-1: bde1bd390e74d55d202f34047c22f303c631c8b5<br>SHA-256: 8e7235833e8de1da0089569abc8754cf561bcd1bc31e98ca09b7aef1949f3109 |
| 2.572 | Released: Jul 7, 2026<br>SHA-1: 680ecd36004256c1b8f948d9fbff6e80bbd92fe6<br>SHA-256: 0c3ea851b126ee86a921822a3b99635497 53e629a52d7a4e3b3378e3d5e23f26 |
| 2.571 | Released: Jun 30, 2026<br>SHA-1: f782160fcf62147d835edc00d2c3dbdadb424b23<br>SHA-256: 3b9100b27639b1465c43cabe73acda51c2e78424fb8114fd499cd970f667d02e |
| 2.570 | Released: Jun 23, 2026<br>SHA-1: 0f423bd3baefaf8f33ffe62d14efd633674a61e1<br>SHA-256: 64404d9b406dba0f6276791393748dddff6944c2d8d82afaef3a86d9667eeceb |
| 2.569 | Released: Jun 16, 2026<br>SHA-1: adb1898475c3d1f01101d7fd0dc58db54cc55181<br>SHA-256: 9aa4df8f1d6515931bf25b07b86f2fc706a94c955dc1cbb39c21b8fa50f27c4e |
| 2.568.1 | Released: Jul 9, 2026<br>SHA-1: 89be8e837d5fb39d7350fccce8c58711797c1499<br>SHA-256: 58f24f3965fbef7708629fbe158d51bf138ffd577cadbc86b46367e8ad0beb83 |

3. Launch command prompt on your system and give the following command

```
C:\Users\15L>java -jar "C:\Program Files\Jenkins\jenkins.war"
Running from: C:\Program Files\Jenkins\Jenkins.war
webroot: C:\Users\15L\.jenkins\war
2026-08-03 10:04:42.127+0000 [id=1]     INFO    winstone.Logger#logInternal: Beginning extraction from war file
2026-08-03 10:04:43.210+0000 [id=1]     WARNING o.e.j.ee9.nested.ContextHandler#setContextPath: Empty contextPath
2026-08-03 10:04:43.249+0000 [id=1]     INFO    org.eclipse.jetty.server.Server#doStart: jetty-12.1.5; built: 2025-12-03T22:18:24.732Z; git: 4905d09d7b6801e792d7b73946cf7b66e4be25d6; jvm 21.0.2+13-LTS-58
2026-08-03 10:04:43.257+0000 [id=1]     INFO    org.eclipse.jetty.server.Server#doStop: Stopped oejs.Server@54eb2b70{STOPPING}[12.1.5,sto=0]
2026-08-03 10:04:43.257+0000 [id=1]     INFO    winstone.Logger#logInternal: Jetty shutdown successfully
java.io.IOException: Failed to start Jetty
        at Jenkins Main ClassLoader//winstone.Launcher.<init>(Launcher.java:197)
        at Jenkins Main ClassLoader//winstone.Launcher.main(Launcher.java:493)
        at java.base/jdk.internal.reflect.DirectMethodHandleAccessor.invoke(DirectMethodHandleAccessor.java:103)
        at java.base/java.lang.reflect.Method.invoke(Method.java:580)
        at executable.Main.main(Main.java:332)
Caused by: java.io.IOException: Failed to bind to 0.0.0.0/0.0.0.0:8080
        at Jenkins Main ClassLoader//org.eclipse.jetty.server.ServerConnector.openAcceptChannel(ServerConnector.java:349)
        at Jenkins Main ClassLoader//org.eclipse.jetty.server.ServerConnector.open(ServerConnector.java:313)
        at Jenkins Main ClassLoader//org.eclipse.jetty.server.Server.lambda$doStart$0(Server.java:617)
        at java.base/java.util.stream.ForEachOps$ForEachOp$OfRef.accept(ForEachOps.java:184)
        at java.base/java.util.stream.ReferencePipeline$3$1.accept(ReferencePipeline.java:197)
        at java.base/java.util.stream.ReferencePipeline$2$1.accept(ReferencePipeline.java:179)
        at java.base/java.util.Spliterators$ArraySpliterator.forEachRemaining(Spliterators.java:1024)
        at java.base/java.util.stream.AbstractPipeline.copyInto(AbstractPipeline.java:509)
        at java.base/java.util.stream.AbstractPipeline.wrapAndCopyInto(AbstractPipeline.java:499)
        at java.base/java.util.stream.ForEachOps$ForEachOp.evaluateSequential(ForEachOps.java:151)
        at java.base/java.util.stream.ForEachOps$ForEachOp$OfRef.evaluateSequential(ForEachOps.java:174)
        at java.base/java.util.stream.AbstractPipeline.evaluate(AbstractPipeline.java:234)
        at java.base/java.util.stream.ReferencePipeline.forEach(ReferencePipeline.java:596)
        at Jenkins Main ClassLoader//org.eclipse.jetty.server.Server.doStart(Server.java:613)
        at Jenkins Main ClassLoader//org.eclipse.jetty.util.component.AbstractLifeCycle.start(AbstractLifeCycle.java:92)
        at Jenkins Main ClassLoader//winstone.Launcher.<init>(Launcher.java:193)
        ... 4 more
Caused by: java.net.BindException: Address already in use: bind
        at java.base/sun.nio.ch.Net.bind0(Native Method)
        at java.base/sun.nio.ch.Net.bind(Net.java:565)
        at java.base/sun.nio.ch.ServerSocketChannelImpl.netBind(ServerSocketChannelImpl.java:344)
        at java.base/sun.nio.ch.ServerSocketChannelImpl.bind(ServerSocketChannelImpl.java:301)
        at Jenkins Main ClassLoader//org.eclipse.jetty.server.ServerConnector.openAcceptChannel(ServerConnector.java:344)
        ... 19 more
2026-08-03 10:04:43.258+0000 [id=1]     SEVERE  winstone.Logger#logInternal: Container startup failed
java.net.BindException: Address already in use: bind
        at java.base/sun.nio.ch.Net.bind0(Native Method)
        at java.base/sun.nio.ch.Net.bind(Net.java:565)
        at java.base/sun.nio.ch.ServerSocketChannelImpl.netBind(ServerSocketChannelImpl.java:344)
        at java.base/sun.nio.ch.ServerSocketChannelImpl.bind(ServerSocketChannelImpl.java:301)
        at Jenkins Main ClassLoader//org.eclipse.jetty.server.ServerConnector.openAcceptChannel(ServerConnector.java:344)
Caused: java.io.IOException: Failed to bind to 0.0.0.0/0.0.0.0:8080
        at Jenkins Main ClassLoader//org.eclipse.jetty.server.ServerConnector.openAcceptChannel(ServerConnector.java:349)
```

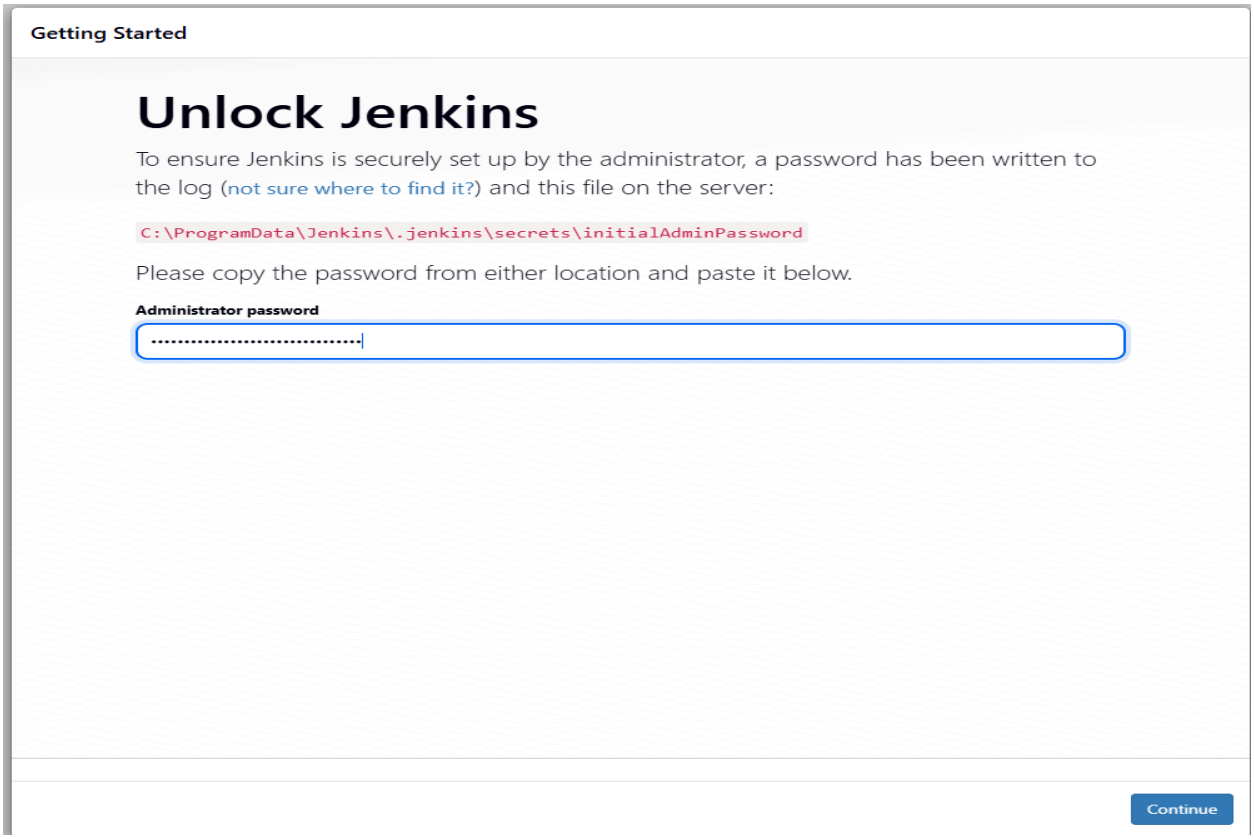
4. Navigate to the folder

```
        at executable.Main.main(Main.java:332)

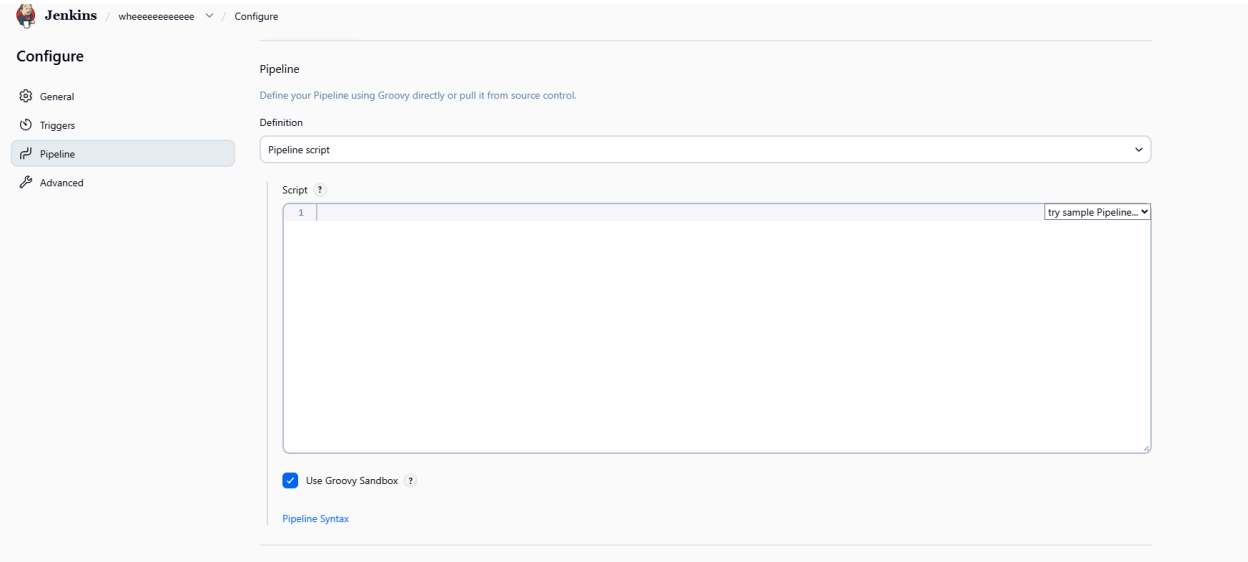
C:\Users\15L>cd "C:\Program Files\Jenkins"

C:\Program Files\Jenkins>java -jar jenkins.war
Running from: C:\Program Files\Jenkins\Jenkins.war
webroot: C:\Users\15L\.jenkins\war
2026-08-03 10:07:11.878+0000 [id=1]     INFO    winstone.Logger#logInternal: Beginning extr
2026-08-03 10:07:11.941+0000 [id=1]     WARNING o.e.j.ee9.nested.ContextHandler#setContextP
2026-08-03 10:07:12.020+0000 [id=1]     INFO    org.eclipse.jetty.server.Server#doStart: je
2026-08-03 10:07:12.030+0000 [id=1]     INFO    org.eclipse.jetty.server.Server#doStop: Sto
2026-08-03 10:07:12.031+0000 [id=1]     INFO    winstone.Logger#logInternal: Jetty shutdown
java.io.IOException: Failed to start Jetty
        at Jenkins Main ClassLoader//winstone.Launcher.<init>(Launcher.java:197)
        at Jenkins Main ClassLoader//winstone.Launcher.main(Launcher.java:493)
        at java.base/jdk.internal.reflect.DirectMethodHandleAccessor.invoke(DirectMethodHan
        at java.base/java.lang.reflect.Method.invoke(Method.java:580)
        at executable.Main.main(Main.java:332)
```

5. Start Jenkins: Run the command C:\Program Files\Jenkins>java -jar jenkins.war to start Jenkins server.
6. Access Web UI: open localhost 8090




7. Open jenkins and pipeline

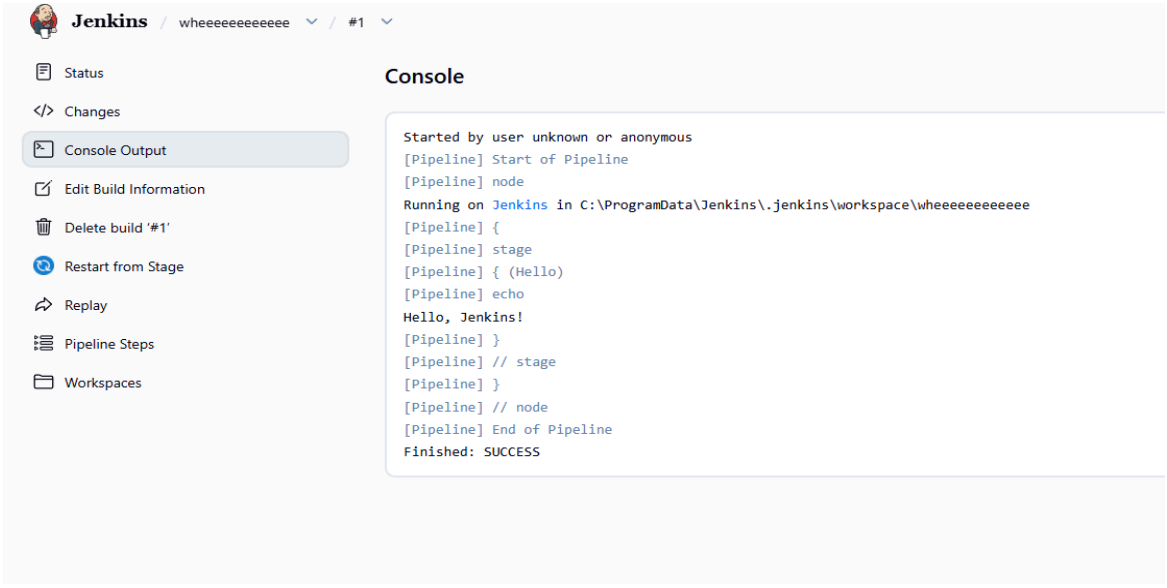



Code:

```
pipeline {
    agent any

    stages {
        stage('Hello') {
            steps {
                echo 'Hello, Jenkins!'
            }
        }
    }
}
```

Output:



Conclusion:

Jenkins Pipeline automates the software development process by integrating stages like build, test, and deployment to support Continuous Integration and Continuous Delivery. In this experiment, Jenkins was successfully installed and configured by verifying the JDK, launching Jenkins using the jenkins.war file, and accessing the setup page through a browser. A Declarative Pipeline was created and executed, demonstrating how Jenkins automates tasks and improves the efficiency and reliability of the CI/CD process.